

Section 5A. Validation Set

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Recap: Training Error versus Testing Error

Assume a *training* dataset $\mathcal{D}_{\text{Tr}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, a *test* dataset $\mathcal{D}_{\text{Te}}$, and a parametric model for regression $f(\mathbf{x}; \theta)$, where θ is a vector of parameters (e.g., β in a linear regression)

- ▶ We learn the parameters of our model solving the following optimization

$$\hat{\theta} = \arg \min_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Tr}}} (y_i - f(\mathbf{x}_i; \theta))^2$$

- ▶ The **training error** is the average error your model presents in its training, i.e.,

$$\text{MSE}_{\text{Tr}} = \frac{1}{|\mathcal{D}_{\text{Tr}}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Tr}}} \left(y_i - f(\mathbf{x}_i; \hat{\theta}) \right)^2$$

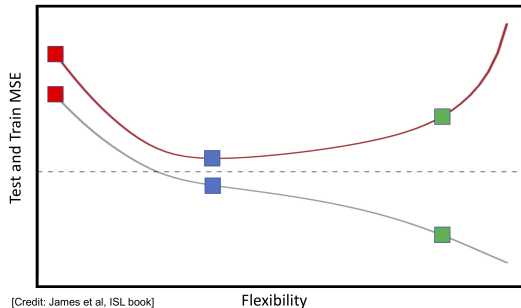
- ▶ The **testing error** is the average error we expect while predicting the output of a new observation, one not used in training. We can estimate this error using the test dataset:

$$\text{MSE}_{\text{Te}} = \frac{1}{|\mathcal{D}_{\text{Te}}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Te}}} \left(y_i - f(\mathbf{x}_i; \hat{\theta}) \right)^2$$

Choosing the Right Model

In practice, we can consider a family of parametric models, $f_1(\mathbf{x}; \theta_1), f_2(\mathbf{x}; \theta_1), \dots, f_D(\mathbf{x}; \theta_D)$, with different levels of flexibility (e.g., polynomial functions of different degrees).

Objective: We want to choose a model with the level of flexibility minimizing the *testing error* (not the training error)...



Estimating the Test Error

In practice, we are only given one dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ containing labeled data. How can we estimate the testing error from a single dataset?

► To estimate the testing error, we can use *resampling methods*:

1. **Validation Set Approach**
2. **K-Fold Cross-Validation**
3. **The Bootstrap** (also used to estimate the standard errors of our parameters)

Main Idea: Resampling methods estimate the test error by holding a subset of the training observations and use them later on for testing

Validation Set (VS) Approach

VS is used to estimate the test MSE of a parametric model $f(\mathbf{x}; \theta)$

Main Steps: Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, take the following steps:

1. Randomly partition the available data samples into two subsets: A *training subset* $\mathcal{D}_{\text{Ts}}$ and a *validation subset* $\mathcal{D}_{\text{Vs}}$
2. Train the model using the data on the training subset:

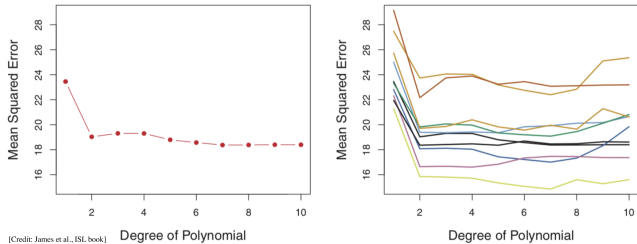
$$\hat{\theta}_{\text{Tr}} = \arg \min_{\theta} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Ts}}} (y_i - f(\mathbf{x}_i; \theta))^2$$

3. Use the fitted model to predict outputs in the *validation subset* and compute the error:

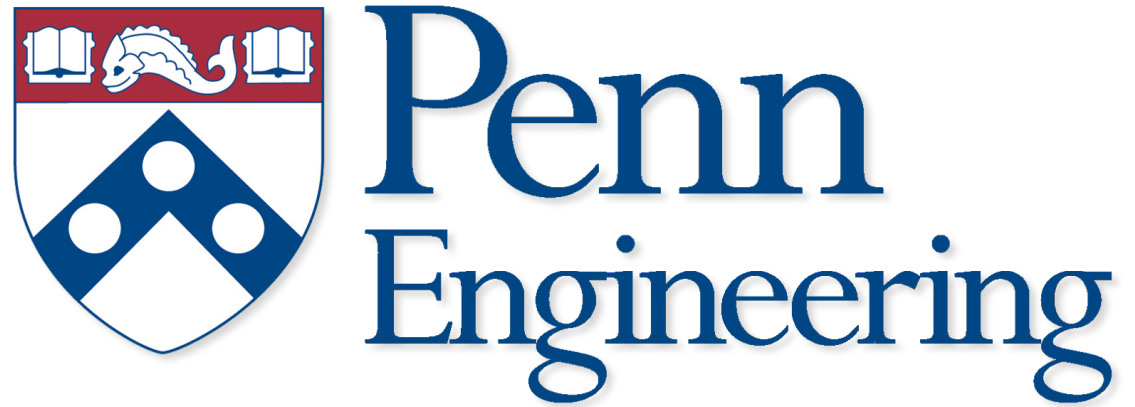
$$\text{MSE}_{\text{Vs}} = \frac{1}{|\mathcal{D}_{\text{Vs}}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{Vs}}} \left(y_i - f(\mathbf{x}_i; \hat{\theta}_{\text{Tr}}) \right)^2$$

Validation Set: Example

Use the VS approach to estimate the test MSE of a family of polynomial models in the Automobile dataset (Fig. one random split (left) vs. 10 random splits (right))



- ▶ **Drawbacks** of the validation set approach:
 - ▶ The MSE_{VS} estimate varies highly with how the dataset is split
 - ▶ Only half the observations are used in training (maybe not enough)
 - ▶ The MSE_{VS} tends to overestimate the test MSE



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